

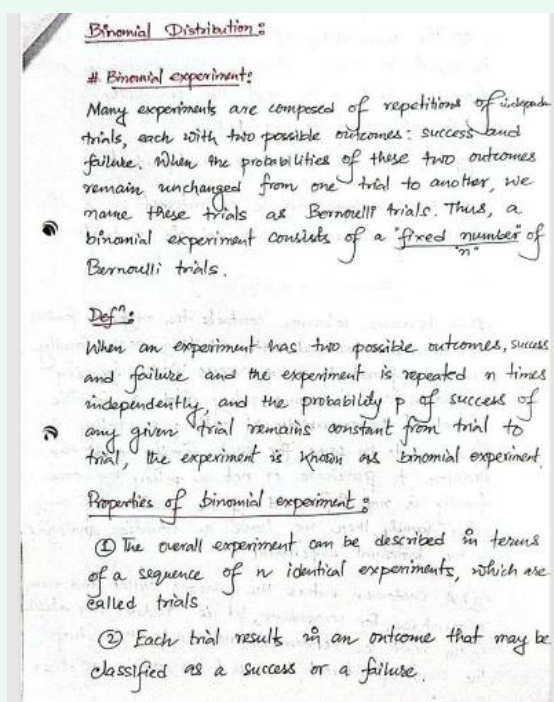
ChatGPT - Math Final



I will give you screenshots of some handwritten math notes. You will explain me that topic in the way it is in the note and give intuition for better understanding. If it is a math or derivation, you will explain it step by step clearly. OK?

Of course!  Please upload the screenshots of your handwritten math notes, and I'll carefully explain them step by step exactly as they appear in your notes, while also giving you the intuition so you understand not just *how* but also *why* the math works.

Go ahead and share the first one!



Great! Let's carefully go through your handwritten notes step by step.

Topic: Binomial Distribution

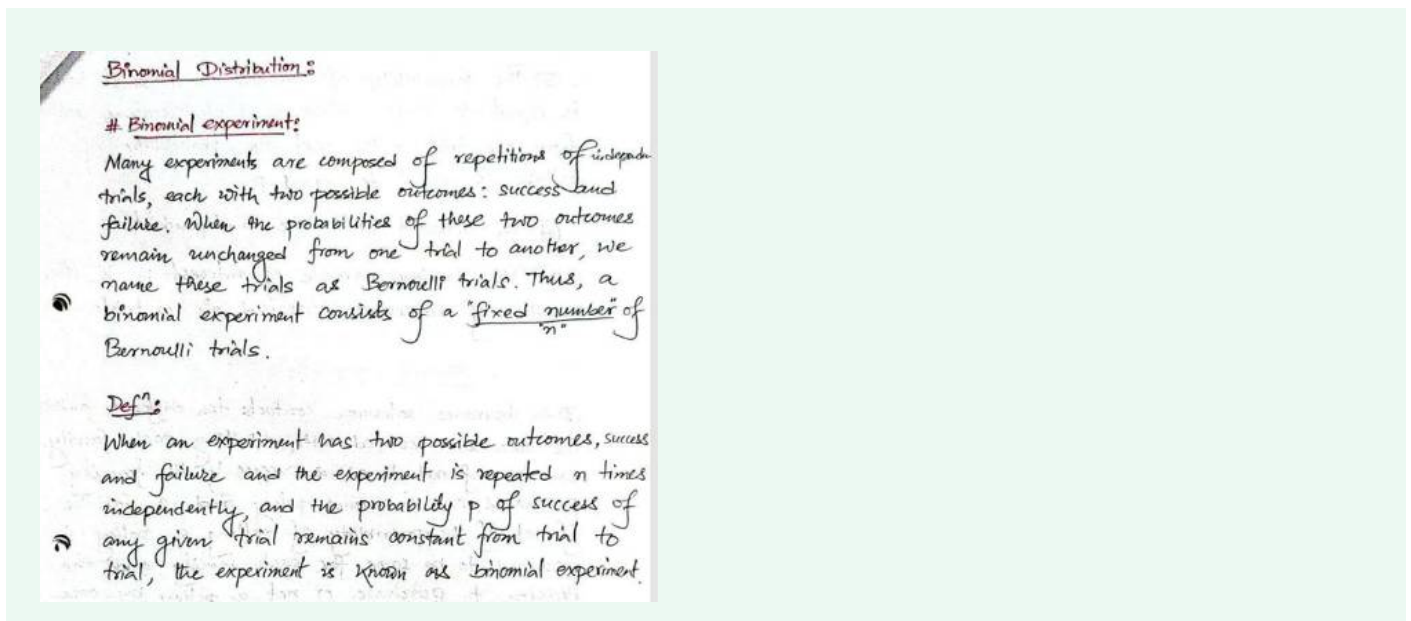
1. Binomial Experiment

- The notes start by saying:

Many experiments are composed of repetitions of independent trials, each with two possible outcomes: success and failure.

👉 This means we are looking at experiments that are repeated several times, where every single trial can only give **two outcomes**:

- **Success** (e.g., tossing a coin → getting heads)
- **Failure** (e.g., tossing a coin → not heads i.e., tails)
- If the **probabilities remain the same** in each trial,



Perfect 👍 Let's carefully explain this part from your notes.

Binomial Distribution – Binomial Experiment

✦ Explanation

- Many experiments consist of **repeated independent trials**.
- Each trial has **two possible outcomes**:
 - **Success**
 - **Failure**

👉 Example: tossing a coin → Head (success) or Tail (failure).

- If the **probability of success and failure stays the same** for each trial, these are called **Bernoulli trials**.
- A **binomial experiment** is then defined as:
A set of a **fixed number (n)** of Bernoulli trials.

♦ Definition (from your notes)

*When an experiment has two possible outcomes (success and failure) and the experiment is repeated **n times independently**, and the probability **p** of success of any trial remains constant from trial to trial, the experiment is known as a **binomial experiment**.*

♦ Intuition

Think of it like this:

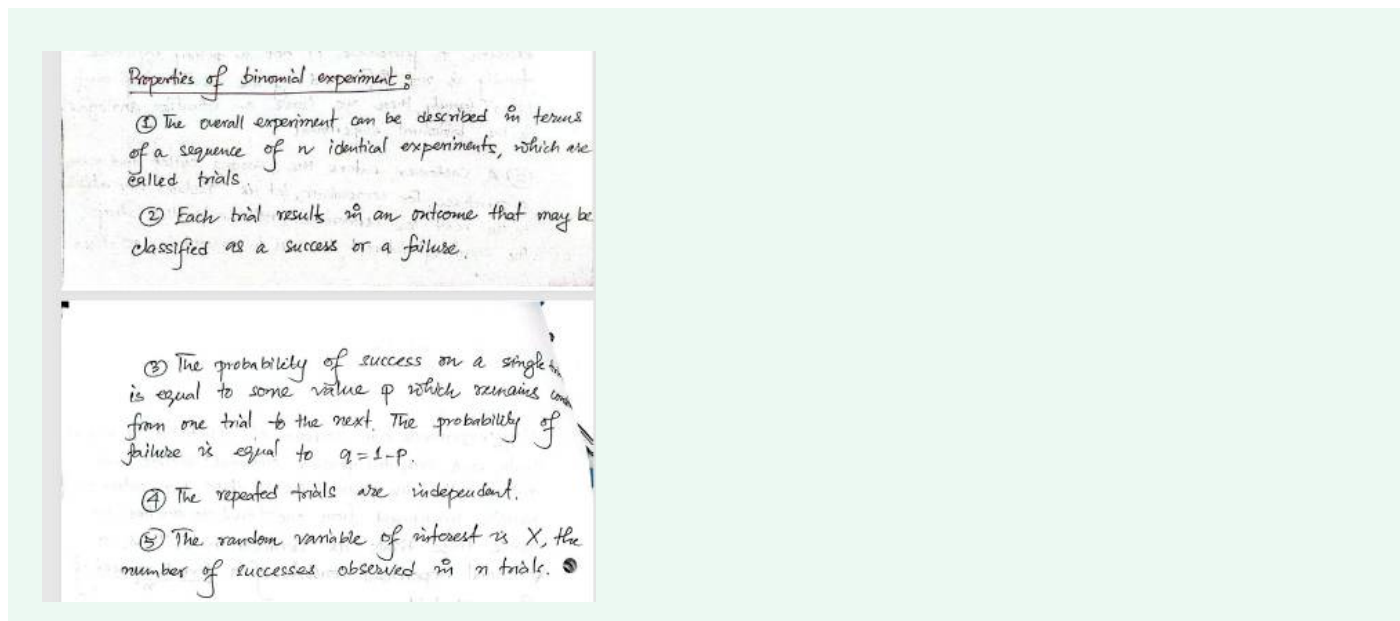
- You're playing a game with a coin (two outcomes).
- You flip it multiple times (say, 10 flips).
- Each flip is independent (one flip doesn't affect the next).
- The probability of success (say, getting heads, $p = 0.5$) remains the same every time.

That whole setup is what we call a **binomial experiment**.

✅ So in summary:

- **Trials** = Bernoulli trials (success/failure).
- **n** = fixed number of trials.
- **p** = constant probability of success.
- **Independent** = outcomes don't influence each other.

Do you want me to also **go through the "Properties of binomial experiment"** part from your first image step by step?



Perfect 🙌 Now let's break down the **Properties of Binomial Experiment** from your notes step by step with explanations and intuition.

Properties of a Binomial Experiment

① The experiment has a fixed number of trials (n).

- The overall experiment is described as a sequence of **n identical experiments** (called trials).
- Example: Tossing a coin **10 times** → here $n = 10$.

👉 The key is that **n is fixed in advance**.

② Each trial has only two outcomes.

- Each trial results in either:
 - **Success** (e.g., head in coin toss, correct answer, etc.)
 - **Failure** (e.g., tail in coin toss, wrong answer, etc.)

👉 So no “third option” exists. This makes it a **binary outcome**.

③ The probability of success (p) is constant.

- Each trial has the same probability **p** of success.
- The probability of failure is **q = 1 - p**.

👉 Example: Tossing a fair coin always gives $p = 0.5$ (success = head), $q = 0.5$ (failure = tail).

④ The trials are independent.

- The outcome of one trial does **not** affect the outcome of another.
- Example: If you toss a coin once and get heads, it does not change the chance of getting heads in the next toss.

👉 This independence is crucial; otherwise, it wouldn't be binomial.

⑤ The random variable of interest is **X = number of successes**.

- We are not interested in the detailed sequence of results, but in **how many successes** occurred in n trials.
- Example: Toss a coin 5 times $\rightarrow X =$ number of heads (could be 0,1,2,3,4,5).

👉 So the binomial distribution tells us the **probability of getting X successes in n trials**.

✅ Summary intuition:

A binomial experiment is like asking:

*If I repeat a yes/no experiment (success/failure) **n times**, with success probability **p** each time, what's the probability of getting exactly X successes?*

Would you like me to now **derive the probability formula of the binomial distribution** ($P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$) step by step, just like in your notes?



Examples of Binomial experiment:

① An insurance salesman contacts ten different families. The outcome associated with visiting each family can be referred to as a success if the family purchases an insurance policy and a failure, if not. If the probability of selling a policy is assumed to be same for each family, and the decision to purchase or not a policy by one family is not influenced by the decision of any other family, then we have a situation analogous to the binomial experiment.

② A customer enters the Aarong outlet and makes a purchase. For convenience, let us restrict our attention to the next ten customers who enter the shop. The event of interest here is to ascertain whether

or not any one of the customers makes a purchase. The process can be viewed as a binomial experiment because of the reason that (i) the experiment can be described as a sequence of ten identical trials, one trial for each of the ten customers that will enter the store, (ii) the probability of purchase (p) may be assumed to be the same for all customers

③ An experiment consists in selecting five radios at random from a lot and inspecting and then classifying them as defective and non-defective. A defective radio is labeled a success, while a non-defective radio is labeled a failure. Suppose 20% of the radios are defective. If this probability is assumed to remain constant from trial to trial, then X , the number of successes is a binomial random variable assuming the values 0, 1, 2, 3, 4, and 5. Clearly, the experiment is a binomial experiment with $n=5$ and $p=0.20$.

④ If five cards are drawn in succession from an ordinary deck and each trial is labeled a success

or failure depending on whether the card is a red or black. If each card is replaced and the deck is well shuffled before the next draw is made, then the repeated drawing (trials) are independent and the probability of success (drawing a red card) here remains constant from trial to trial. This is a binomial experiment.

In each of the above experiments, there exists a variable that denotes the number of successes in n -Bernoulli trials. This variable is known as the binomial random variable. A binomial random variable may thus be viewed as a set of values representing the number of successes generated through some binomial experiment.

Excellent 🙌 Now let's carefully go through the **Examples of Binomial Experiments** from your notes.

Examples of Binomial Experiment

① Insurance salesman example

- A salesman contacts **10 families**.
- Each family either:
 - Buys insurance (**success**), or
 - Doesn't buy insurance (**failure**).
- Probability of selling (p) is assumed constant for each family.
- Families' decisions are independent.

👉 Therefore, this setup is a **binomial experiment with $n = 10$ trials**.

② Customer purchase example

- Consider the **next 10 customers** entering Aarong.
- Each customer either:
 - Makes a purchase (**success**), or
 - Doesn't (**failure**).
- Probability of purchase is the same for each customer.
- Decisions of customers are independent.

👉 So again, this is a **binomial experiment with $n = 10$** .

③ Defective radios example

- Pick **5 radios** at random from a lot.
- Each radio is classified as:
 - **Defective (success)**, or
 - **Non-defective (failure)**.

- Suppose 20% are defective $\rightarrow p = 0.20$.
- Trials are independent, probability remains constant.

👉 Thus, this is a **binomial experiment with $n = 5$, $p = 0.20$** .

④ Drawing cards example

- Draw **5 cards** from a deck, where each card is labeled success/failure based on being red/black.
- After each draw, the card is **replaced and deck reshuffled** $\rightarrow$ keeps trials independent.
- Probability of red card (success) remains constant $p = 0.5$.

👉 Hence, this is a **binomial experiment with $n = 5$, $p = 0.5$** .

Conclusion from all examples

- In all these experiments, the **random variable** of interest is:
 $X =$ number of successes in n trials.

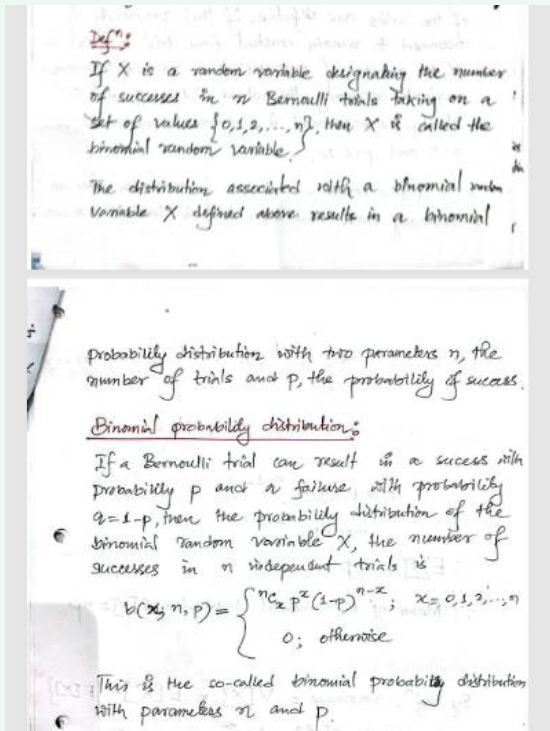
👉 This random variable is called the **Binomial Random Variable**.

✨ Intuition:

A binomial random variable is basically **counting how many times "success" happens** in a fixed number of independent trials.

Would you like me to now move on to the **derivation of the Binomial Probability Formula** $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$ step by step (like in your notes)?





Perfect 👍 Now we've reached the **definition of binomial random variable** and the **binomial probability distribution** in your notes. Let's go step by step.

Definition

- If **X** is a random variable denoting the **number of successes in n Bernoulli trials**,
then X can take values from:

$$X \in \{0, 1, 2, \dots, n\}$$

- This X is called the **Binomial Random Variable**.

👉 Example: If you toss a coin 5 times ($n = 5$), then X = number of heads can be 0, 1, 2, 3, 4, or 5.

Distribution

The distribution associated with this random variable is the **Binomial Probability Distribution**.

It depends on two parameters:

- n = number of trials
- p = probability of success

Binomial Probability Distribution Formula

If each trial can result in:

- Success with probability p
- Failure with probability $q = 1 - p$

Then, the probability that there are exactly x successes in n trials is:

$$b(x; n, p) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x}, & x = 0, 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

Step-by-step intuition of formula

1. Pick which trials are successes

- Out of n trials, choose x to be successful $\rightarrow \binom{n}{x}$ ways.

2. Multiply the probabilities

- Each success has probability p . For x successes $\rightarrow p^x$.
- Each failure has probability $1 - p$. For $(n-x)$ failures $\rightarrow (1 - p)^{n-x}$.

3. Combine them

- Total probability =

$$\binom{n}{x} p^x (1 - p)^{n-x}$$

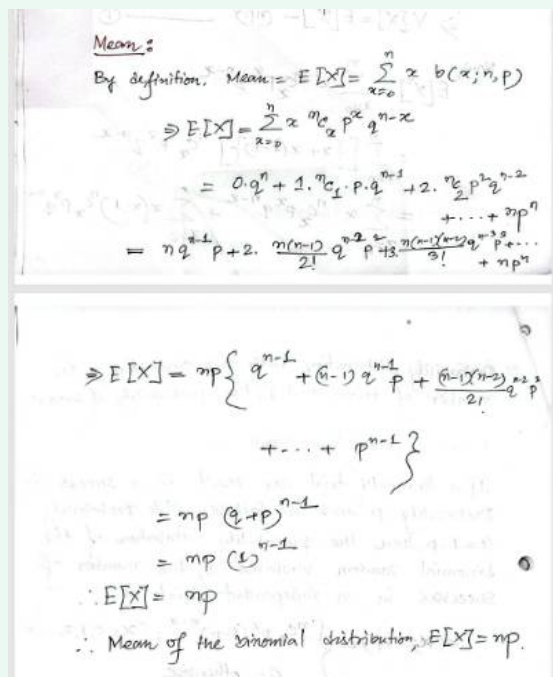
👉 This is why the binomial distribution is also called a **discrete probability distribution** – it gives probabilities only for integers $x = 0, 1, 2, \dots, n$.

✅ So this formula tells us:

What is the probability of getting exactly x successes in n independent trials, where each trial has probability p of success?

Would you like me to **derive this formula step by step with a small example** (like coin toss with $n=3$) so it's crystal clear?





Handwritten derivation of the mean of a binomial distribution:

$$\begin{aligned} \text{Mean:} \\ \text{By definition, Mean} &= E[X] = \sum_{x=0}^n x b(x; n, p) \\ &\Rightarrow E[X] = \sum_{x=0}^n x \binom{n}{x} p^x q^{n-x} \\ &= 0 \cdot q^n + 1 \cdot \binom{n}{1} p \cdot q^{n-1} + 2 \cdot \binom{n}{2} p^2 q^{n-2} + \dots + n p^n \\ &= n q^{n-1} p + 2 \cdot \frac{n(n-1)}{2!} q^{n-2} p^2 + \dots + n p^n \\ &\Rightarrow E[X] = n p \left\{ q^{n-1} + (n-1) q^{n-2} p + \frac{(n-1)(n-2)}{2!} q^{n-3} p^2 + \dots + p^{n-1} \right\} \\ &= n p (q+p)^{n-1} \\ &= n p (1)^{n-1} \\ &\therefore E[X] = n p \\ &\therefore \text{Mean of the binomial distribution, } E[X] = n p. \end{aligned}$$

show it exactly how its done in the note

Perfect 👍 Now let's carefully go through the **derivation of the mean of a binomial distribution** exactly as it is written in your notes, step by step.

Mean of Binomial Distribution

Step 1: Definition

By definition,

$$\text{Mean} = E[X] = \sum_{x=0}^n x b(x; n, p)$$

where

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad q = 1 - p$$

Step 2: Substitute distribution

$$E[X] = \sum_{x=0}^n x \binom{n}{x} p^x q^{n-x}$$

Step 3: Expand terms

$$= 0 \cdot q^n + 1 \cdot \binom{n}{1} p q^{n-1} + 2 \cdot \binom{n}{2} p^2 q^{n-2} + \dots + n \cdot \binom{n}{n} p^n$$

Step 4: Simplify coefficients

Notice the pattern:

$$= n q^{n-1} p + \frac{n(n-1)}{2!} q^{n-2} p^2 + \frac{n(n-1)(n-2)}{3!} q^{n-3} p^3 + \dots + n p^n$$

Step 5: Factor out np

$$E[X] = np \left\{ q^{n-1} + \frac{(n-1)}{1!} q^{n-2} p + \frac{(n-1)(n-2)}{2!} q^{n-3} p^2 + \dots + p^{n-1} \right\}$$

Step 6: Recognize binomial expansion

The bracket is actually the expansion of:

$$(q + p)^{n-1}$$

Step 7: Simplify

Since $q + p = 1$,

$$E[X] = np(1)^{n-1} = np$$

✓ Final Result

$E[X] = np$

So, the **mean of the binomial distribution** is:

$$\mu = np$$

Would you like me to also **show the derivation of the variance** from your notes in the same detailed, step-by-step way?



Variance:

By defⁿ, variance = $V[X] = E[X^2] - \{E[X]\}^2$

$\Rightarrow V[X] = E[X^2] - (np)^2$ ——— (1)

Now, $E[X^2] = \sum_{x=0}^n x^2 \binom{n}{x} p^x q^{n-x}$

$= \sum_{x=0}^n [x + x(x-1)] \binom{n}{x} p^x q^{n-x}$

$= \sum_{x=0}^n x \binom{n}{x} p^x q^{n-x} + \sum_{x=0}^n x(x-1) \binom{n}{x} p^x q^{n-x}$

$= np + n(n-1)p^2 \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} q^{n-x}$

$= np + n(n-1)p^2 \left\{ \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} q^{n-x} \right\}$

$= np + n(n-1)p^2 (q+p)^{n-2}$

$= np + n(n-1)p^2 (1)$

$= np + n(n-1)p^2$

$= np [1 + (n-1)p]$

$= np [1 + np - p] = np [1 - p + np]$

$\Rightarrow E[X^2] = np [1 - p + np]$

(1) $\Rightarrow V[X] = np [1 - p + np] - n^2 p^2$

$= np [1 - p + np] - n^2 p^2$

$\therefore V[X] = npq$

$\therefore$ Variance of the binomial distribution, $V[X] = npq$

same for it

Awesome—let's do the **variance derivation** exactly the way it appears in your notes, line by line, with the same tricks.

Variance of a Binomial $X \sim \text{Bin}(n, p)$

1) Start from the definition

$$V[X] = E[X^2] - \{E[X]\}^2 \quad (1)$$

We already proved $E[X] = np$.

2) Compute $E[X^2]$

$$E[X^2] = \sum_{x=0}^n x^2 \binom{n}{x} p^x q^{n-x}, \quad q = 1 - p$$

Use the algebraic split $x^2 = x + x(x-1)$:

$$E[X^2] = \sum_{x=0}^n x \binom{n}{x} p^x q^{n-x} + \sum_{x=0}^n x(x-1) \binom{n}{x} p^x q^{n-x}$$

3) First sum (becomes the mean)

$$\sum_{x=0}^n x \binom{n}{x} p^x q^{n-x} = np$$

4) Second sum (use the combinatorial identity)

Use $x(x-1) \binom{n}{x} = n(n-1) \binom{n-2}{x-2}$. Then

$$\sum_{x=0}^n x(x-1) \binom{n}{x} p^x q^{n-x} = n(n-1) p^2 \sum_{x=0}^n \binom{n-2}{x-2} p^{x-2} q^{(n-2)-(x-2)}$$

Shift the index (starts effectively at $x=2$) and recognize a binomial expansion:

$$\sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} q^{(n-2)-(x-2)} = (p+q)^{n-2} = 1$$

Hence

$$\sum_{x=0}^n x(x-1) \binom{n}{x} p^x q^{n-x} = n(n-1)p^2$$

5) Put them together

$$E[X^2] = np + n(n-1)p^2$$

6) Plug into (1) and simplify

$$\begin{aligned} V[X] &= (np + n(n-1)p^2) - (np)^2 \\ &= np + n(n-1)p^2 - n^2p^2 \\ &= np - np^2 \\ &= np(1-p) \\ &= \boxed{npq} \end{aligned}$$

✓ Final result (as in your notes)

- **Mean:** $E[X] = np$
- **Variance:** $V[X] = npq$

If you want, send the next page (MGF, mode, examples, etc.), and I'll do it in the same style.